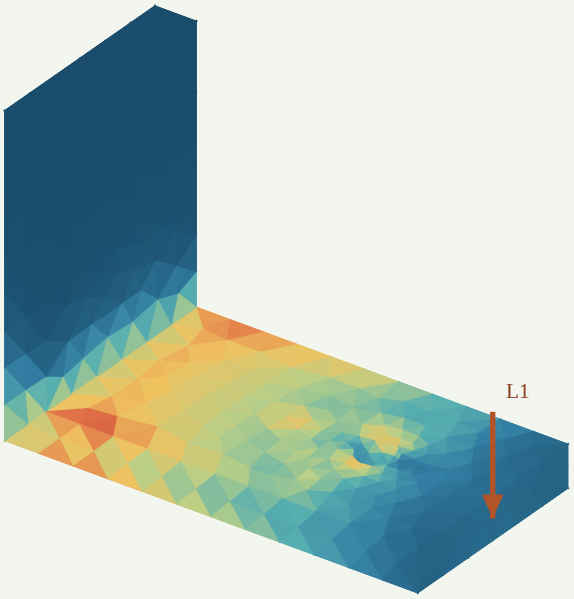
Synthetic demo mounting bracket

Run af02566402df43b2b0125ff688188e74 | 2026-09-09T06:26:01Z

PEAK VON MISES	MAX DISPLACEMENT	YIELD RATIO
14.66 MPa	0.1194 mm	16.37

VON MISES STRESS / MPa



Undeformed geometry. Nodal-averaged colors.



Interpretation

This is a linear-elastic engineering estimate. Peak stress is the maximum unaveraged quadrature-point von Mises stress. The diagram uses volume-weighted nodal averages on undeformed geometry.

A single mesh does not establish convergence or physical acceptance. Yield ratio uses the supplied material strength and is not a safety certification.

Force equilibrium error: 4.899e-09 N. Free-DOF relative residual: 7.737e-09.

Simulation definition

Model	Synthetic demo mounting bracket
Analysis	3D small-strain static linear elasticity
Compute device	apple · metal-3
Solver method	WebGPU conjugate gradient, symmetric Jacobi scaling
Precision	GPU float32 with browser float64 iterative refinement
Material	Aluminum · illustrative
Young modulus	69000 MPa
Poisson ratio	0.33
Supplied yield strength	240 MPa
Density	2700 kg/m3
Geometry source	image_proposal
CAD volume	41436 mm3
Mesh volume	41465 mm3
Estimated mass	111.88 g
Mesh	3,070 tetrahedra / 16,974 quadratic DOFs
Nominal mesh size	7 mm
Minimum element quality	0.44356 (minSICN)
Fixed faces	1 (XYZ translations = 0)

Boundary conditions and verification

L1: face 9, total XYZ force [0.0, 0.0, -100.0] N, distributed uniformly over 320 mm2.

Support reaction XYZ: [6.801883500884287e-10, 1.5352332738416408e-09, 100.00000000460233] N. Moment equilibrium error: 2.582e-07 N mm.

Strain energy: 5.9346 N mm.

Application notes

Synthetic demonstration only. Rear X=0 is fully fixed; total [0,0,-100] N is uniformly distributed on X=80. Bores are unloaded. No fillets, contact, bolts or preload. Check actual dimensions, material condition and mesh convergence before real use.

Assumptions and provenance

- Single homogeneous isotropic solid; quadratic tetrahedral small-strain, linear-elastic static analysis.

- Fixtures are ideal fully fixed faces. Forces are distributed uniformly across the selected mesh faces in global XYZ.

- No contact, bolt preload, plasticity, buckling, fatigue, thermal effects or manufacturing anisotropy.

- Material presets are illustrative inputs. Verify properties for the actual alloy, condition and manufacturing process.

- Peak stress is the maximum unaveraged quadrature-point von Mises stress; display colors are volume-weighted nodal averages.

- Sharp corners and fixture edges can have stress singularities. A single mesh does not establish stress convergence.

- Yield ratio is a linear-elastic indicator, not a safety certification. Physical acceptance requires independent validation.

- Geometry originated from an image proposal. User confirmation does not constitute metrology.

- The image is synthetic; all dimensions are supplied demonstration specifications in mm.

- 90° bracket: base occupies X=0–80, Y=0–40, Z=0–8; upright occupies X=0–8, Y=0–40, Z=0–60. Height is overall.

- Both bores are plain Ø8 mm through-holes: base center X=57.6, Y=20, axis Z; upright center Y=20, Z=43.2, axis X.

- No hidden features or fillets; the legs form one connected solid.

- 7 mm is the initial mesh size. Sharp internal-corner and fixture-edge stresses may be mesh-dependent; assess mesh convergence.

- Material properties are illustrative, not certified. Analysis assumes homogeneous isotropic small-strain linear elasticity and a static load.

Reproducibility

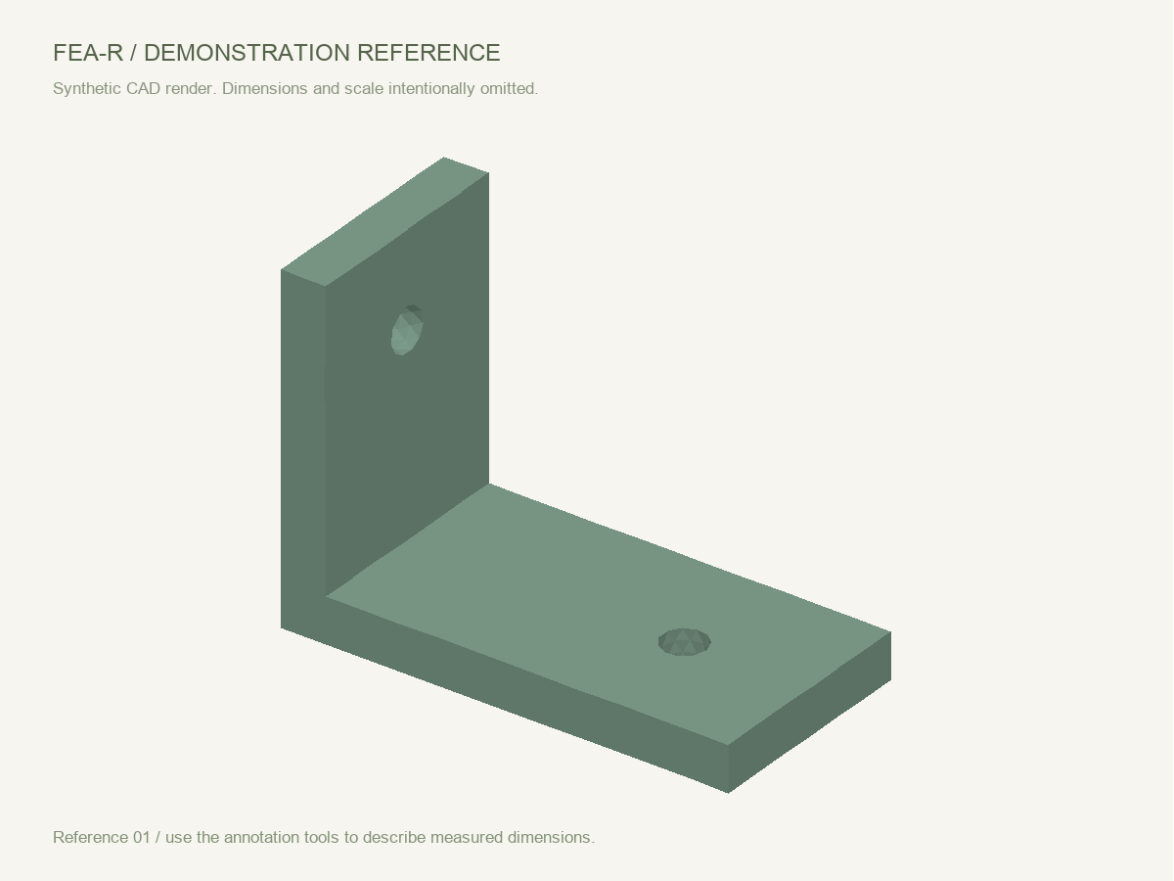
CAD revision: f5f0cb3a22034e0f8b67f1b4733b5573. Solver versions: {'scikit-fem': '12.0.2', 'numpy': '2.5.3', 'scipy': '1.18.1', 'gmsh': '4.15.2'}. Runtime: 19.3 s. No mesh-convergence study was performed for this run.

CAD SHA-256: 5bc45148dc0f8a60c548801dc2b4b9c3ca9986734975082c2832dadafbc65b90

Mesh SHA-256: 1da55f7acc8ba3e491c71453e8718d242710fbe766f073c66400adfd885737b0

The exported package includes the CAD, mesh, inputs, results, quadratic solution arrays, nodal CSV, solver source and pinned dependencies. Material values are supplied inputs, not certified data.

Reference evidence



Annotation 1 (dimension): 80 mm overall base length (supplied demo specification)

Normalized image coordinates: [0.23424369747899154, 0.7241115196078431] to [0.6081932773109243, 0.9201899509803922]